

Project Title: Telecom Customer Churn Analysis

Milestone 4: Product Dashboard & Retention Strategy

Objective:

The objective of this project is to develop a data-driven churn prediction model that identifies customers at risk of leaving the telecom service. Using advanced feature engineering and logistic regression, the model estimates churn probability, enables customer risk scoring, and supports targeted retention strategies. The ultimate goal is to reduce churn rate, protect customer lifetime value (CLV), and improve overall business profitability through proactive intervention.

Tools Used:

Microsoft Excel (Pivot Tables, Statistical Tests, Visualizations), PowerBi

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Section 1: Executive Summary

The company is currently experiencing a 26.54% churn rate, which is significantly higher than the industry benchmark of approximately 20%. This gap represents substantial revenue leakage and long-term customer value erosion.

Comprehensive churn analysis was conducted using descriptive analytics, engagement metrics, and predictive modeling. Key churn drivers identified include:

- Month-to-month contracts
- Manual payment methods (Electronic Check)
- Low service engagement
- Early tenure customers (<12 months)
- High monthly charges without bundled value

A Logistic Regression model was developed to predict churn probability and segment customers into risk tiers. While the current model performance (AUC = 0.483) indicates scope for further optimization, it enables structured customer risk segmentation.

Financial modeling shows:

- Customers Identified (75% recall assumption): ~1,372
- Customers Saved (40% success): ~549
- Revenue Saved: ₹21.94 Lakhs
- Program Cost: ₹6.86 Lakhs
- Net Benefit: ₹15.09 Lakhs
- ROI: 220%

This strategy shifts retention from reactive to proactive, data-driven intervention.

Section 2: Churn Analysis Insights

High-Risk Segments

1. Month-to-Month Contracts – Highest churn contribution
2. Tenure < 12 months – Early-stage churn spike (>50% in first 3 months)
3. Electronic Check Payment Users – High churn correlation
4. Low Engagement Customers (0–2 services)
5. High Monthly Charges (>₹80 segment)

Protective Factors

1. Two-Year Contracts – Very low churn (~5–8%)
2. Automatic Payment Methods – Lower churn risk
3. High Service Engagement (5+ services)
4. Family Customers (Partner/Dependents)
5. Tenure > 24 months – Stabilized churn (<10%)

Core Churn Drivers Identified

- Contract flexibility (lack of commitment)
- Payment friction (manual payment)
- Low switching cost (few services)
- Price sensitivity without perceived value
- Weak early onboarding

Churn risk is highest within the first 6 months and declines sharply after 24 months.

Section 3: Predictive Churn Model

Model Overview

- Model Type: Logistic Regression
- Features Used: 15+ engineered predictors
- Train/Test Split: 70/30 (Stratified)
- Dataset: 7,043 customers
- Class Distribution: 73.46% retained, 26.54% churned

Model Performance (Threshold = 0.7)

- Accuracy: ~54%
- Recall: 29%
- Precision: ~22%
- AUC: 0.483 (Poor discrimination)

The model currently provides segmentation support but requires refinement for production-level deployment.

Risk Scoring Segmentation

Churn Risk Score = Churn Probability × 100

Risk Tier	Score	Range	Action
High Risk	70–100	Immediate retention intervention	
Medium Risk	40–70	Proactive engagement	
Low Risk	0–40	Upsell & loyalty focus	

Revenue at risk is concentrated within high and medium tiers.

Section 4: Differentiated Retention Strategies

Strategy A: High-Risk Intervention

Target: Month-to-month, <12 months tenure, manual payment, low services

Actions:

1. Personalized retention call
2. 20% discount for 1-year contract
3. Free add-on service trial (Online Security / Tech Support)
4. ₹200 incentive for switching to automatic payment

Expected Outcome:

- 40% retention success
- ~549 customers saved
- Revenue saved: ₹21.94 Lakhs
- Cost: ₹6.86 Lakhs
- ROI: 220%

Strategy B: Medium-Risk Engagement

Target: Contract ending customers, declining engagement

Actions:

- Early renewal discount (15–20%)
- Engagement email campaign
- Usage education
- NPS surveys & dissatisfaction triggers

Expected retention success: ~30%

Strategy C: Low-Risk Value Maximization

Target: Long-tenure, high-engagement customers

Actions:

- Loyalty program
- Upsell bundles
- Referral incentives
- Anniversary recognition

Goal: Increase ARPU by 10–15%

Section 5: Implementation Roadmap

Phase 1 (Months 1–2)

- Improve predictive model
- Integrate scoring into CRM
- Train retention team
- Deploy automated risk alerts

Phase 2 (Month 3)

- Pilot high-risk strategy (20% sample)
- Measure conversion impact

Phase 3 (Months 4–6)

- Full rollout
- Launch medium-risk campaigns
- Monitor weekly KPIs

Phase 4 (Months 7–12)

- A/B test offers
- Optimize thresholds
- Improve model features

Total Year 1 Investment: ₹45 Lakhs

Section 6: Expected Business Impact

Churn Reduction

- Current churn: 26.54%
- Target churn: 20%
- Projected reduction: ~4–6%

Financial Impact

- Revenue Saved: ₹21.94 Lakhs
- Net Benefit: ₹15.09 Lakhs
- ROI: 220%

Operational Benefits

- Reduced customer acquisition pressure
 - Improved customer lifetime value
 - Higher engagement & contract conversion
 - Better forecasting accuracy
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Conclusion

Although the predictive model requires further optimization (AUC improvement), structured segmentation and targeted retention strategies already demonstrate positive ROI.

The organization should focus on:

- Contract conversion
- Payment automation
- Early tenure engagement
- Service bundling

By combining analytics with proactive intervention, churn can be systematically reduced while increasing long-term customer value.
